ETL vs ELT

Most Data Engineering Beginners Confuse These Two!

ETL

Extract → Transform → Load

ELT

Extract → Load → Transform

☆ The order changes **EVERYTHING!**

- ☑ Two different approaches
- ☑ Different processing locations
- ☑ Different use cases
- ☑ Impact on performance, cost and scalability

ETL

(Extract → Transform → Load)

1 EXTRACT

Collect data from multiple sources in different formats.

Sources: Databases, APIs, Files, Applications, Logs, SaaS

2 TRANSFORM

Clean, validate and convert data into the required structure.

Tasks: Clean → Filter → Join → Aggregate → Validate → Enrich

3 LOAD

Load the transformed data into the destination system.

Targets: Data Warehouse, Data Marts, Analytics Databases

In ETL, data is TRANSFORMED before it reaches the destination.

Common Tools

Airflow · AWS Glue · Talend · Informatica · Hadoop · SSIS

ELT

(Extract → Load → Transform)

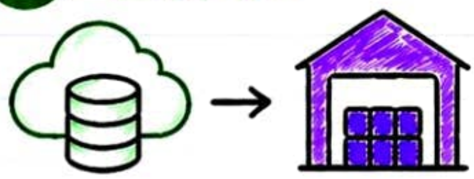
⭐ Load BEFORE Transform

1 EXTRACT

Collect raw data from different sources in its original format.

Sources: Databases, APIs, Files, Applications, Logs, SaaS

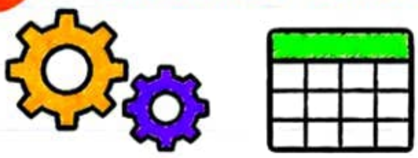
2 LOAD



Load the raw data directly into the target data platform.

Targets: Cloud Data Warehouse, Data Lake, Lakehouse

3 TRANSFORM



Transform the data inside the target platform when needed.

Tasks: Clean → Join → Aggregate → Validate → Enrich → Model

ELT leverages the power of the target platform to process large volumes of raw data efficiently.

Common Tools

Airflow | Apache Spark | dbt | Snowflake | BigQuery | Redshift

⭐ In ELT, raw data is stored first, transformed later!

ETL vs ELT

Key Differences at a Glance

ASPECT	ETL	ELT
Order	Extract → Transform → Load (Transform happens first)	Extract → Load → Transform (Load happens first)
Where Transformation Happens	Outside the target system (ETL Tool / Processing Engine)	Inside the target system (Data Warehouse / Cloud Platform)
Data Loaded into Target	Cleaned, transformed and prepared data	Raw data first, transformed later when required
Handling Large Volumes	Heavy transformations take more time and resources before loading	Fast loading of large volumes, transformations done on scalable platform
Flexibility	Less flexible once data is transformed before loading	More flexible, multiple transformations can be done on the same data
Use Cases	Traditional systems, Small to medium datasets, Strict data quality rules	Modern cloud analytics, Big data, Data exploration, Raw data retention
Cost	Higher upfront cost (ETL Tools, Infra)	Lower upfront cost, Pay as you use in cloud
Examples / Tools	Informatica, Talend, SSIS, Datastage	Snowflake, BigQuery, Redshift, dbt, Spark SQL

Quick Summary

ETL
Transform first, then Load.

VS

ELT
Load first, then Transform.

Both are great – choose based on your architecture, data volume, business needs and platform.

ETL vs ELT

— Real-World Example —

Let's say an e-commerce company generates data from

Orders | Payments | Customers | Products | Website Events

Many Sources
Many Formats
Huge Volume!

ETL Approach

Transform before loading into the Data Warehouse

Extract from Sources → Transform (Clean, Join, Aggregate, Validate) → Load Prepared Data into Data Warehouse → Analytics & Reporting

☆ Only processed and cleaned data is stored in the target.

ELT Approach

Load first, then transform inside the target platform

Extract from Sources → Load Raw Data into Data Lake / Data Warehouse → Transform inside Target Platform (dbt, SQL, Spark, Stored Procedures) → Analytics & Reporting

☆ Raw data is preserved and transformed as needed.

Common Tools →

Apache Airflow | Apache Spark | dbt | Snowflake | BigQuery | Redshift | Amazon S3

KEY TAKEAWAY

★ ETL = Sources → Transform → Data Warehouse

☆ ELT = Sources → Data Warehouse → Transform

ELT leverages the scalability and compute power of modern cloud data platforms.

ETL vs ELT

When to Use Which?

Choose ETL when:

- Data must be cleaned and validated before reaching the target.
- Only transformed / prepared data should be stored in the target.
- Working with traditional or legacy systems.
- Strict business rules, data quality and compliance are required.
- Smaller datasets where pre-processing is practical and efficient.

Choose ELT when:

- Using modern cloud data platforms (Snowflake, BigQuery, Redshift etc.).
- Handling large volumes of data.
- Need to preserve raw data for future use.
- Different teams need different transformations.
- Want faster ingestion and cost-effective processing.

INTERVIEW TAKEAWAY

Q: What is the main difference between ETL and ELT?

A: ETL transforms data before loading it into the target system.
ELT loads the data first and performs transformations inside the target platform.

Easy Way to Remember

ETL = Transform BEFORE Load

ELT = Load BEFORE Transform

There is no one-size-fits-all! Choose based on your data, use case, platform and business needs.

SAVE THIS POST for your next **DATA ENGINEERING INTERVIEW!**